

TECHNICAL REQUIREMENTS DOCUMENT (TRD)

System Architecture

Three-tier architecture: React frontend, Node.js + Express backend, MongoDB database.

Frontend Design

React with Bootstrap UI, video homepage, dynamic filters, maps integration, responsive design.

Backend Design

REST API using Express.js, JWT authentication, role-based access (customer/owner), booking validation logic.

Database Schema

Collections: Users (roles), Restaurants (photos, menu, details), Reservations (slots, guests), Reviews.

Key Functional Modules

Authentication module, restaurant module, booking module, review module, payment simulation module.

API Structure

Auth APIs, Restaurant APIs, Booking APIs, Review APIs, Owner dashboard APIs.

Security

JWT authentication, password hashing, protected routes, input validation.

Performance

Efficient queries, optimized API calls, fast UI rendering.

Integration

Google Maps API (for location), media storage (images/videos), frontend-backend via REST APIs.

Deployment

Frontend on Vercel, Backend on Render, Database on MongoDB Atlas.